

Euclid's Division Lemma

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1 Statement

Given $j > 0$ and a divisor $k > 0$, there exists a unique solution (q, r) with quotient $q > 0$ and remainder $r > 0$ to the equation $j/k = q + r/k$ given the condition $0 \leq r < k$, where we say q is the quotient and r is the remainder.

2 Proof

Consider the sequence $0k < 1k < 2k < 3k < 4k < 5k < \dots$

Either $j = qk$ for some q (case 1)

or

$qk < j < (q+1)k$ for some q . (case 2)

We see that the value of q is unique.

In the first case $j = qk + 0$ meaning $j/k = q + r/k$ where $r = 0$.

In the second case $j = qk + r$ and since j is somewhere between an integer multiple of k , then $0 \leq r < k$, meaning $j/k = q + r/k$ where $0 \leq r < k$.